EE 474 Introduction to Control Systems Homework 8

Due Wednesday November 11, 2026

Reading:

- Franklin *et al.*: Chapter 5, Beginning through Section 5.3 (6th, 7th or 8th ed.)

1 Homework :

A) PI and PD Control

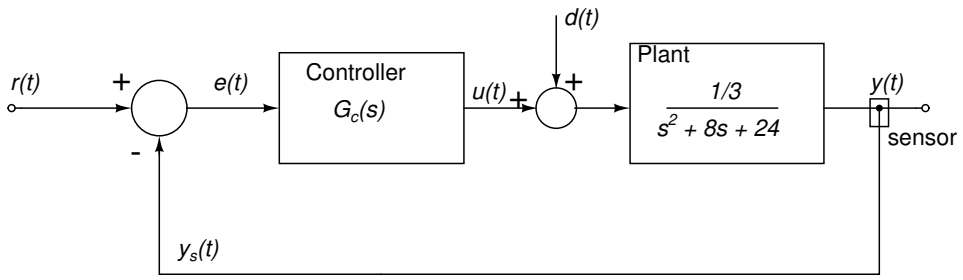


Figure 1: A system with PI-type control.

A.1) PI Control, use

$$G_c(s) = \frac{k_p s + k_i}{s} \tag{1}$$

with $k_p = 10$, $k_i = 200$.

A.1.a) Determine the transfer functions:

i) $T_{ry}(s)$	The input-output transfer function
ii) $T_{re}(s)$	The error transfer function
iii) $T_{dy}(s)$	The disturbance transfer function

A.1.b) Draw a pole-zero diagram of $T_{ry}(s)$, label the dominant mode.

A.2) Based on the dominant mode pole locations, determine

i) t_p, the peak time
ii) ρ, the damping
iii) M_p, the percent overshoot
iv) t_r, the rise time
v) t_s, the settling time.

A.3) Analytically determine e_{ss}, the Steady State Error, by determining the system type and K_{Bode} .

A.4) Setup $T_{ry}(s)$ in Matlab, run and plot the (true) step response. Determine these values for the complete system

i) t_p, the peak time
ii) ρ, the damping
iii) M_p, the percent overshoot
iv) t_r, the rise time
v) t_s, the settling time
vi) e_{ss}, the steady state error

A.5) (Essay question) What is the value of the methods of steps A.1.b-A.2, relative to the method of step A.4.

A.6) Based on parts A.1.b-A.4, which measures based on the dominant modes came out most accurate ?

Which measures based on the dominant modes (part 3) came out least accurate ?

B) PD control, use

$$G_c(s) = k_d s + k_p \tag{2}$$

with $k_p = 300$, and k_d to be determined.

B.1) Experiment with k_d

B.1.a) Derive $T_{ry}(s)$, the closed loop transfer function with PD control, and with k_p and k_d as variables.

Set $k_p = 300$ and determine the poles locations for several values of k_d on the range from 0.5 to 10.

B.1.b) Hand sketch a plot of the closed-loop pole locations as a function of k_d .

Suggestion: use Matlab to do all the computation. You could write:

```
Kds = [0.5 1 2 3 4 5 6 7 8 9 10 ]
kk = 0;
for kd = Kds, kk=kk+1,
  ...
  Try = (Form the closed-loop transfer function)
  Poles(:, kk) = pole(Try);
  ...
  Rho(kk) = compute rho for each value of kd
end
```

B.1.c) Select a value of k_d to give $\rho = 0.4$, be rather accurate, e.g., $0.41 < \rho < 0.39$.

Using your value of kd chosen to give rho = 0.4:

B.2) What is the dominant mode of this system ?
(Same as asking "where are the dominant pole(s)")

B.3) Analyze your system (analytically, based on the dominant mode) to determine:

i) t_p, the peak time
ii) ρ, the damping
iii) M_p, the percent overshoot
iv) t_r, the rise time
v) t_s, the settling time.

B.4) Analytically determine e_{ss}, the Steady State Error, by determining the system type and K_{Bode} .

B.5) Analyze the step response of your system (step response, full system, you can use the step and stepinfo commands) to determine:

i) t_p, the peak time
ii) ρ, the damping
iii) M_p, the percent overshoot
iv) t_r, the rise time
v) t_s, the settling time
vi) e_{ss}, the steady state error (`not indicated by stepinfo()`)

B.6) Which of the values based on dominate mode are closest to their analysis values ? Which are farthest ?

C) Compare the PI and PD controllers

Consider these parameters, the desirable characteristic is indicated.

Controller performance measure	Desirable characteristic
i) t_p, the peak time	Fast
ii) ρ, the damping	Near one
iii) M_p, the percent overshoot	Near zero
iv) t_r, the rise time	Fast
v) t_s, the settling time.	Fast
vi) e_{ss}, the steady-state error	Small

C.1) What characteristic is most enhanced by the PI control system ?

C.2) What characteristic is most enhanced by the PD control system (tuned for ρ = 0.4) ?